The Geometry of Quantization

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Chapter 1

The Harmonic Oscillator

1.1 Classical Setting

Classically, the harmonic oscillator is described by a 2nd order ODE, called Newton's Second Law

$$m\ddot{x} = -kx$$

in the 'configuration space' $\mathbb{R} = \{x \mid x \in \mathbb{R}\}$. This 2nd order ODE can be turned into a system of 1st order ODEs as follows: First, define the canonical variables as

$$q := x \qquad p := m\dot{x}$$

Then, we obtain

$$\dot{q} = p/m \qquad \dot{p} = -kx$$

These equations define a flow in the "phase space"

$$\mathbb{R}^2 = \{(q,p) \mid q,p \in \mathbb{R}\}$$

If we define the Hamiltonian (energy) function $H : \mathbb{R}^2 \to \mathbb{R}$ as

$$H(q,p) = \frac{p^2}{2m} + \frac{kq^2}{2}$$

we get

$$\dot{q} = \frac{\partial H}{\partial p} \qquad \dot{p} = -\frac{\partial H}{\partial q}$$

the so-called Hamilton's equations.

This system of equations has two important properties:

- H is constant along integral curves:

$$\frac{dH}{dt} = \frac{\partial H}{\partial q}\dot{q} + \frac{\partial H}{\partial p}\dot{p} = \frac{\partial H}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial H}{\partial p}\frac{\partial H}{\partial q} = 0$$

  Thus, the integral curves lie in the level sets

$$H^{-1}(E) = \{(q,p) \mid p^2/2m + kq^2/2 = E\}$$

  which is an ellipse.

- The flow preserves the area in the phase space. To see this, notice that our vector field, called the Hamiltonian vector field, is divergenceless:

$$X_H = \left(\frac{\partial H}{\partial p}, -\frac{\partial H}{\partial q}\right) \to \nabla X_H = \frac{\partial^2 H}{\partial q \partial p} - \frac{\partial^2 H}{\partial p \partial q} = 0$$

1.2 Quantum Setting

Quantum mechanically, the harmonic oscillator is described by a 'wave function' $\psi : \mathbb{R}^2 \to \mathbb{C}$, $\psi(x,t)$ satisfying the Schrödinger equation

$$i\hbar\frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} + \frac{kx^2}{2}\psi$$

If we define the operator

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{kx^2}{2}$$

then we can express the Schrödinger equation as

$$i\hbar\frac{\partial \psi}{\partial t} = \hat{H}\psi$$

If we assume that the solution $\psi(x,t)$ is normalized as

$$\int_{\mathbb{R}} |\psi(x,t)|^2 dx = 1$$

then its square norm can be interpreted as a PDF:

$$\rho(x,t) = |\psi(x,t)|^2 \to \text{ Probability Distribution}$$

For a given t, $\psi(x,t)$ defines a function $\mathbb{R} \to \mathbb{C}$. Thus, $\psi(x,t)$ can be thought of as a path in $C^\infty(\mathbb{R},\mathbb{C})$, and the Schrödinger equation defines a vector field on $C^\infty(\mathbb{R},\mathbb{C})$.

1.3 Quantization and Classical Limit

To connect classical mechanics to quantum mechanics, we can use the substitutions

$$q \mapsto \hat{q} = x \qquad p \mapsto \hat{p} = -i\hbar\frac{\partial}{\partial x}$$

Then, we have the correspondence

$$H = \frac{p^2}{2m} + \frac{kx^2}{2} \mapsto \hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{kx^2}{2}$$

However, if we want to quantize

$$(1+q^2)p^2$$

we must make a choice between

$$(1+\hat{q}^2)\hat{p}^2 \quad \hat{p}^2(1+\hat{q}^2) \quad \ldots$$

since $\hat{p}$ and $\hat{q}$ do not commute. In the other direction, we expect that the classical

solutions to Hamilton's equations are in some sense an approximation to the solutions of the Schrödinger equation. It turns out that such an approximation is a good approximation when $\hbar$ is small enough. Thus, we shouldn't think of $\hbar$ as a fixed number; we should think of it as a formal parameter.

1.4 The Hamilton-Jacobi Equation

Consider a 1-d system with Hamiltonian

$$H(q,p) = \frac{p^2}{2m} + V(q)$$

Then, Hamilton's equations become

$$\dot{q} = p/m \qquad \dot{p} = -V'(q)$$

and the Schrödinger equation becomes

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi$$

To solve the Schrödinger equation, first we look for the separable solutions

$$\psi(x,t) = \varphi(x)e^{-iEt/\hbar}$$

This leads to the eigenvalue problem

$$(\hat{H} - E)\varphi = 0$$

Let's consider the solutions of the form

$$\varphi(x) = e^{iS(x)/\hbar} \Rightarrow \text{ WKB Ansatz}$$

for some $S : \mathbb{R} \to \mathbb{R}$ called the phase function. In this case, we get

$$\frac{S'(x)^2}{2m} + (V(x) - E) - \frac{i\hbar}{2m} S''(x) = 0$$

As a first approximation, we can drop the last term and get

$$H(x, S'(x)) = \frac{S'(x)^2}{2m} + V(x) = E$$

the Hamilton-Jacobi equation. To interpret this equation geometrically, consider the phase space

$$\mathbb{R}^2 \simeq T^*\mathbb{R} = \{(q, p) \mid q, p \in \mathbb{R}\}$$

and define a function called the graph of dS by

$$\iota_{dS} : \mathbb{R} \to T^*\mathbb{R} \qquad x \mapsto (x, S'(x))$$

Then we have

$$H(x, S'(x)) = E \iff H(\iota_{dS}(x)) = E \iff \operatorname{im}(\iota_{dS}) \subseteq H^{-1}(E)$$

Thus, the Hamilton-Jacobi equation says that the image of ι_{dS} lies inside a level set of the Hamiltonian. We can generalize these results to higher dimensions. In $\mathbb{R}^n$, we have

$$H(q, p) = \sum_{i=1}^{n} (p_i^2/2m) + V(q) \to \hat{H} = -\frac{\hbar^2}{2m}\Delta + V$$

If we consider the WKB ansatz $\varphi = e^{iS/\hbar}$, we get

$$\frac{||\nabla S||^2}{2m} + (V - E) - \frac{i\hbar}{2m}\Delta S = 0$$

Thus, in the first order, we have the Hamilton-Jacobi equation

$$\frac{||\nabla S||^2}{2m} + V(x) = H(x, \nabla S) = E$$

In that case, our error is of order $\hbar$, i.e.,

$$(\hat{H} - E)\varphi = O(\hbar)$$

We call a phase function $S : \mathbb{R}^n \to \mathbb{R}$ admissible if it satisfies the Hamilton-Jacobi equation. Then, the image

$$L = \operatorname{im}(\iota_{dS})$$

satisfies

- L is an n-dimensional submanifold of $H^{-1}(E)$.
- The pullback to L of the form

$$\alpha_n = \sum_{j=1}^{n} p_j dq_j$$

  is exact, i.e., we have

$$\alpha_n|_L = \sum_j \frac{\partial S}{\partial q_j} dq_j = dS$$

- The canonical projection $\pi : T^*\mathbb{R}^n \to \mathbb{R}^n$ induces a diffeomorphism $L \simeq \mathbb{R}^n$. Thus, L is projectable.

Of course, these assertions hold when S is globally defined. If it is not, we have the weakened version:

- L is an n-dimensional submanifold of $H^{-1}(E)$.
- The pullback to L of the form

$$\alpha_n = \sum_{j=1}^{n} p_j dq_j$$

  is closed, i.e., we have

$$d\alpha_n|_L := \omega_n|_L = 0$$

In the latter case, we say L is a Lagrangian submanifold of $\mathbb{R}^{2n}$. If L satisfies the former strong assertions, we say L is an exact, projectable Lagrangian submanifold. Example: Consider the harmonic oscillator with $L = H^{-1}(E)$, i.e.,

an ellipse.

- The pullback to L of $\alpha = pdq$ is closed since $\dim L = 1$, but not exact since

$$\int_L pdq = \int dp \wedge dq = \text{Area of ellipse} \neq 0$$

- L fails to project diffeomorphically to $\mathbb{R}$ since the former is compact while the latter is not.

1.5 Hamilton-Jacobi Theorem

Consider the Hamiltonian vector field

$$X_H = \sum_j \left(\frac{\partial H}{\partial p_j} \frac{\partial}{\partial q_j} - \frac{\partial H}{\partial q_j} \frac{\partial}{\partial p_j} \right)$$

satisfying

$$X_H \lrcorner\, d\alpha_n = -dH \quad \text{i.e.,} \quad -d\alpha_n(X_H, -) = dH(-)$$

If the Lagrangian submanifold L lies in a level set of H, then TL lies in the kernel of dH since H is constant on L. Thus, the two-form $d\alpha_n$ vanishes on the subspace

$$\langle T_pL, X_H(p) \rangle \leq T_p\mathbb{R}^{2n}$$

for each $p \in L$. Since $d\alpha_n$ defines a nondegenerate, skew-symmetric bilinear form, by linear algebra, a subspace on which $d\alpha_n$ vanishes can be at most n-dimensional. Thus, we have

$$X_H(p) \in T_pL$$

In other words, X_H is tangent to L, or the restriction of X_H defines a vector field on L. Conversely, assume X_H is tangent to the Lagrangian submanifold L. Then, we have

$$dH|_L(T_pL) = -d\alpha_n|_L(X_H(p), T_pL) = 0 \Rightarrow dH|_L = 0$$

Thus, H is locally constant on L.

A function $H : \mathbb{R}^{2n} \to \mathbb{R}$ is locally constant on a Lagrangian submanifold L of $\mathbb{R}^{2n}$ iff X_H is tangent to L.

1.6 The WKB Approximation

Recall that if $S : \mathbb{R}^n \to \mathbb{R}$ is an admissible function, then $\varphi = e^{iS/\hbar}$ satisfies

$$(\hat{H} - E)\varphi = O(\hbar)$$

To obtain a better approximation, consider

$$\varphi(x) = a(x)e^{iS(x)/\hbar}$$

This gives

$$(\hat{H} - E)\varphi = -\frac{1}{2m}\left[i\hbar\left(a\Delta S + 2\sum_j \frac{\partial a}{\partial x_j}\frac{\partial S}{\partial x_j}\right) + \hbar^2\Delta a\right]e^{iS/\hbar}$$

Thus, as a second approximation we have

$$a\Delta S + 2\sum_j \frac{\partial a}{\partial x_j}\frac{\partial S}{\partial x_j} = 0 \to \text{Transport Equation}$$

Thus, we obtain a solution satisfying

$$(\hat{H} - E)\varphi = O(\hbar^2)$$

Example: In 1-d, we have

$$aS'' + 2a'S' = 0 \Rightarrow a = \frac{c}{\sqrt{S'}} = \frac{c}{[2m(E-V)]^{1/4}}$$

Since a is independent of S, we can use this expression for non-projectable Lagrangian submanifolds as well. However, a is not well-defined. For instance, if $V(x) = x^2$ we have

$$a = \begin{cases} |x| < \sqrt{E} \to \text{Well-defined} \\ |x| = \sqrt{E} \to \text{Singularity} \\ |x| > \sqrt{E} \to \mathbb{C} \end{cases}$$

1.7 Geometry of Transport Equation

Multiplying the transport equation by a gives

$$\sum_j \frac{\partial}{\partial x_j}\left(a^2 \frac{\partial S}{\partial x_j}\right) = 0$$

Our aim is to transform this property to $L = \mathrm{im}(\iota_{dS})$. First, notice

$$H = \sum_i \frac{p_i^2}{2} + V(q) \to X_H|_L = \sum_j \left(\frac{\partial S}{\partial x_j}\frac{\partial}{\partial q_j} - \frac{\partial V}{\partial q_j}\frac{\partial}{\partial p_j}\right)$$

Projecting this to $\mathbb{R}^n$ gives

$$X_H^{(x)} = \sum_j \frac{\partial S}{\partial x_j}\frac{\partial}{\partial q_j} = \nabla S$$

Thus, $a^2 X_H^{(x)}$ is divergenceless for $|dx| = |dx_1 \wedge \cdots \wedge dx_n|$ on $\mathbb{R}^n$:

$$\mathcal{L}_{a^2 X_H^{(x)}}(|dx|) = 0$$

By Cartan's formula, this amounts to

$$\mathcal{L}_{X_H^{(x)}}(a^2|dx|) = 0$$

Since the Lie derivative is natural, this leads to

$$\pi^*(a^2|dx|) \text{ is invariant under the flow of } X_H$$

Thus, the half-density

$$\sqrt{\pi^*(a^2|dx|)} := a$$

is invariant under X_H.

Semiclassical State $\to$ Lagrangian submanifold of $\mathbb{R}^{2n}$ with an invariant half-density a on L

Example: Let $H = (p^2 + q^2)/2$ with

$$L = H^{-1}(E) = \{(\cos\theta, \sin\theta) \mid \theta \in \mathbb{R}/2\pi\mathbb{Z}\}$$

Then, the flow of X_H corresponds to rotation of θ, thus we must have

$$\pi^*(a^2|dx|) = a^2 dq(\theta) = -a^2 \sin\theta d\theta \sim -d\theta \Rightarrow a \sim 1/\sqrt{p}$$

$$\sqrt{d\theta} \to \text{Unique invariant half-density on } L$$

Remarks:

- Notice that the semiclassical states of the H.O. are not singular.
- Our construction does not depend on S, thus it makes sense for arbitrary Lagrangian submanifolds.

As a further step, we identify the quantum states with the Hilbert space

$$\mathcal{H}_{\mathbb{R}^n} = \left\{ \sigma \text{ half-density on } \mathbb{R}^n \;\middle|\; \langle \sigma, \tau \rangle = \int_{\mathbb{R}^n} \sigma \bar{\tau} \right\}$$

and define the Hamiltonian operator as

$$\hat{H}(a\sqrt{|dx|}) = (\hat{H}a)\sqrt{|dx|}$$

Then, the second-order approximate solution to the eigenvalue problem $\hat{H} I_\hbar = E I_\hbar$ is given by

$$I_\hbar = [(dS)^* a]\, e^{iS/\hbar} \in \mathcal{H}_{\mathbb{R}^n}$$

Chapter 2

Symplectic Geometry

2.1 Symplectic Vector Spaces

Let V be an m-dimensional vector space over $\mathbb{R}$ endowed with a bilinear form

$$\omega : V \times V \to \mathbb{R}$$

This gives a map $\tilde{\omega}$ defined by contraction:

$$\tilde{\omega} : V \to V^* \qquad x \mapsto \omega(x, -)$$

The ω-orthogonal of a subspace $W \leq V$ is defined by

$$W^\perp = \{x \in V \mid \omega(x, W) = 0\}$$

The bilinear form ω is called a symplectic form if

- ω is nondegenerate $\iff$ $\tilde{\omega}$ is an isomorphism $\iff$ $V^\perp = 0$
- $\omega(x, y) = -\omega(y, x)$

(Notice that these two properties imply $m = 2n$) In that case, we say (V, ω) is a symplectic vector space. We can also define the symmetries of a symplectic vector space, i.e., symplectomorphisms, as follows:

$$Sp(V) = \{A \in GL(V) \mid \omega(Ax, Ay) = \omega(x, y)\}$$

Canonical Example: Let E be a real n-dimensional vector space and $V = E \oplus E^*$. Define $\omega : V \times V \to \mathbb{R}$ as

$$\omega((x, \lambda), (x', \lambda')) = \lambda'(x) - \lambda(x')$$

2.2 Lagrangian Subspaces

The ω-orthogonal of a subspace W of a symplectic vector space (V, ω) is called the symplectic orthogonal of W. We have

- $W^{\perp\perp} = W$
- $\dim W^{\perp} = \dim V - \dim W$
- $A \leq B \Rightarrow B^{\perp} \leq A^{\perp}$
- $(A + B)^{\perp} = A^{\perp} \cap B^{\perp}$
- $(A \cap B)^{\perp} = A^{\perp} + B^{\perp}$

A subspace $L \leq V$ is called a Lagrangian subspace provided

- $\omega|_L = 0$
- $\dim L = \frac{1}{2} \dim V$

Let me sketch some results without proofs:

Theorem 2.2.1. *Every symplectic vector space has a Lagrangian subspace.*

Theorem 2.2.2. *Every $2n$-dimensional symplectic vector space is symplectomorphic to $(\mathbb{R}^{2n}, \omega_n)$ where*

$$\omega_n = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$$

2.3 The Lagrangian Grassmannian

Let V be a $2n$-dimensional symplectic vector space. The Lagrangian Grassmannian $\mathcal{L}(V)$ is defined to be

$$\mathcal{L}(V) = \{\text{Lagrangian subspaces of } V\}$$

This set has a natural manifold structure, and in fact, we have

$$\mathcal{L}(V) \simeq U(n)/O(n)$$

In particular, the fundamental group of $\mathcal{L}(V)$ (i.e., the group of closed loops (up to homotopy) in $\mathcal{L}(V)$) is isomorphic to that of S^1:

$$\pi_1(\mathcal{L}(V)) \simeq \pi_1(S^1) \simeq \mathbb{Z}$$

Thus, we can define the 'winding number' of a closed loop in $\mathcal{L}(V)$, giving the cohomology element

$$\mu_V \in H^1(\mathcal{L}(V), \mathbb{Z})$$

Example: Consider $\mathbb{R}^2$ and the closed loop

$$L(t) = \langle \cos t, \sin t \rangle \in \mathcal{L}(\mathbb{R}^2) \qquad t \in [0, \pi]$$

This loop performs one rotation, so the corresponding winding number is 1.

2.4 Symplectic Manifolds

To motivate the definition of a symplectic manifold, let's recall the $\mathbb{R}^{2n}$ case. First, define the differential forms

$$\alpha_n = \sum_{i=1}^{n} p_i dq_i \qquad \omega_n = -d\alpha_n = \sum_{i=1}^{n} dq_i \wedge dp_i$$

The 2-form ω_n defines a symplectic vector space structure on each tangent space of $\mathbb{R}^{2n}$. The dynamics of the system are determined by the Hamiltonian vector field

$$X_H = \sum_{i=1}^{n} \left(\frac{\partial H}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial H}{\partial q_i} \frac{\partial}{\partial p_i} \right)$$

This vector field is uniquely determined by the relation

$$i_{X_H} \omega_n = dH \Rightarrow X_H = \tilde{\omega}_n^{-1}(dH)$$

where we used the non-degeneracy of ω_n. Thus, the symplectic form ω_n completely determines Hamilton's equations. This description leads us to the following:

- Energy Conservation

  $$\mathcal{L}_{X_H} H = X_H H = dH(X_H) = \omega_n(X_H, X_H) = 0$$

  where we used the 2-form (antisymmetric) nature of ω_n.

- Preservation of Symplectic Area

  $$\mathcal{L}_{X_H} \omega_n = d(i_{X_H} \omega_n) + i_{X_H} d\omega_n = 0$$

  where we used the closedness of ω_n.

So, let's make the following definition: A symplectic structure on a manifold M is a closed, nondegenerate 2-form ω on M. In this case, we say (M, ω) is a symplectic manifold.

Example: $(\mathbb{R}^{2n}, \omega_n = \sum_{i=1}^{n} dq_i \wedge dp_i)$

2.5 Maslov Index

A symplectic vector bundle is a vector bundle $E \to B$ where each fiber is a symplectic vector space. The Lagrangian subbundle is defined similarly. Now, let's define the famous Maslov index: Suppose we are given the manifold $\mathbb{R}^{2n} \simeq T^*\mathbb{R}^n$ and a Lagrangian submanifold L. Let $\gamma(t)$ be a closed loop in L. By means of the Gauss map

$$G : L \to \mathcal{L}(\mathbb{R}^{2n}) \qquad p \mapsto T_pL$$

we can get a closed loop in $\mathcal{L}(\mathbb{R}^{2n})$. Now, define the Maslov index $\mu_L(\gamma(t))$ of the closed loop $\gamma(t)$ to be the winding number of this closed loop in $\mathcal{L}(\mathbb{R}^{2n})$. Example: In $\mathbb{R}^2$ consider $L = S^1$ and

$$\gamma(t) = (\cos t, \sin t) \quad t \in [0, 2\pi]$$

When t moves from 0 to 2π, the Lagrangian subspace $T_{\gamma(t)}L$ performs two rotations, so

$$\mu_L(\gamma(t)) = 2$$

2.6 Cotangent Bundles

The most important examples of symplectic manifolds are cotangent bundles. Let M be a smooth manifold. Recall that T_p^*M denotes the vector space of covectors at $p \in M$. Now, the cotangent bundle T^*M is defined to be

$$T^*M = \{(x, b) \mid x \in M, b \in T_x^*M\}$$

We have the natural projection map

$$\pi : T^*M \to M \qquad \pi(x, b) = x$$

Locally, T^*M can be parametrized by $2n$ coordinates $(q_1, \ldots, q_n; p_1, \ldots, p_n)$ where the q's stand for coordinates of M. Now, define the 1-form α_M on T^*M, called the tautological 1-form as follows:

$$\alpha_M|_{(x,b)}(v) = b(\pi_* v)$$

Locally, this form can be expressed as

$$\alpha_M = \sum_{i=1}^{n} p_i dq_i$$

The exterior derivative of the tautological 1-form gives us a symplectic form

$$-d\alpha_M = \omega_M = \sum_{i=1}^{n} dq_i \wedge dp_i$$

2.7 The Liouville Class

From now on, we will focus on the Lagrangian submanifolds of the cotangent bundles. As we know, the easiest Lagrangian submanifolds are exact, projectable Lagrangian submanifolds, so let's consider what these concepts mean in our context. For a cotangent bundle T^*M and Lagrangian submanifold $L \subseteq T^*M$, we can consider the projection of L into M:

$$\pi_L := \pi|_L : L \to M$$

If π_L is a diffeomorphism, we say L is a projectable Lagrangian submanifold. Recall that L is an exact Lagrangian if it is the graph of an exact 1-form. So, let's consider graphs of 1-forms. Let φ be a 1-form on M. Its graph is defined as

$$\iota_\varphi : M \to T^*M \qquad x \mapsto (x, \varphi_x)$$

We have the following (trivial) fact about graphs of 1-forms:

- $\iota_\varphi^* \alpha_M = \varphi$

This together with the naturality of the exterior derivative gives

$$d\varphi = -\iota_\varphi^* \omega_M$$

Thus, the projectable Lagrangian submanifolds of T^*M are exactly the graphs of closed 1-forms on M. Now, for φ to be exact, i.e., $\varphi = dS$, we must have

$$\alpha_M|_L = d(S \circ \pi)$$

so $\alpha_M|_L$ must be an exact form. Of course, this is not always the case. However, we always have

$$d\alpha_M|_L = -\omega_M|_L = 0$$

Thus, the 1-form $\alpha_M|_L$ defines the so-called Liouville class

$$\lambda_L \in H^1(L; \mathbb{R})$$

Example: Let $M = \mathbb{R} = \{q \in \mathbb{R}\} \to T^*M = \{(q,p) \in \mathbb{R}^2\}$ and consider the Lagrangian

$$L = S^1 = \{(\cos\theta, \sin\theta)\}$$

As it is well-known, we have an isomorphism $H^1(S^1; \mathbb{R}) \simeq \mathbb{R}$ which is given by integration. Now, the Liouville class of L is given by

$$\lambda_L = \int_L \alpha_M = \int_L p dq = -\pi \in H^1(S^1; \mathbb{R}) \simeq \mathbb{R}$$

Chapter 3

Quantization of Cotangent Bundles

3.1 Mechanics on Manifolds

Let's extend our discussion of classical/quantum mechanics to manifolds. Let M be a configuration space of a physical system and T^*M be the corresponding phase space. The Hamiltonian $H : M \to \mathbb{R}$ is defined to be $H(q,p) = T(q,p) + V(q)$ where

$$T(q,p) = \frac{1}{2}\sum_{ij} g^{ij}(q,p)p_i p_j$$

is the kinetic energy and

$$V \to \mathbb{R}$$

is the potential energy. The case $V = 0$ is especially important in geometry, which gives us the geodesics of M. In quantum mechanics, one defines the Hamiltonian operator $\hat{H} : C^\infty(M) \to C^\infty(M)$ as

$$\hat{H} = -\frac{\hbar^2}{2m}\Delta + V$$

where Δ is the Laplace-Beltrami operator on M. One can also define the corresponding Hamiltonian operator $\hat{H} : \mathcal{H}_M \to \mathcal{H}_M$ as usual. Now, the Schrödinger equation

$$\hat{H}\psi = E\psi$$

has the second-order WKB approximation

$$\psi = e^{iS/\hbar}a$$

where

$$H \circ \iota_{dS} = E \qquad a\Delta S + 2\mathcal{L}_{\nabla S}a = 0$$

Now, let's talk about the geometry of this WKB solution. If $L \subseteq H^{-1}(E)$ is an exact Lagrangian submanifold, i.e.,

- $\pi_L := \pi|_L$ is a diffeomorphism
- $\alpha_M|_L$ is an exact 1-form on L

Let $\phi \to \mathbb{R}$ be a primitive of $\alpha_M|_L$ and a be a half-form on L invariant under the action of $X_H|_L$ (whose existence is guaranteed by the Hamilton-Jacobi theorem), then a semiclassical state on M is given by

$$(\pi_L^{-1})^* e^{i\phi/\hbar} a$$

Thus, a semiclassical state is determined by:

- A Lagrangian submanifold $L \subseteq T^*M$ satisfying $L \subseteq H^{-1}(E)$
- A half-form a satisfying
$$\mathcal{L}_{X_H|_L} a = 0$$

The question we must answer is the following: What will we do if L is not exact (no ϕ) and not projectable (no π_L^{-1})?

3.2 Handling Non-Exactness

Assume L is not an exact Lagrangian, i.e., the 1-form $\alpha_M|_L$ is not exact. It is still closed, however, and thus locally exact:

$$L = \bigcup_j L_j \qquad \exists \phi_j : L_j \to \mathbb{R} \text{ such that } d\phi_j = \alpha_M|_{L_j}$$

Then, we can quantize each L_j as

$$(\pi_{L_j}^{-1})^* e^{i\phi_j/\hbar} a$$

To quantize L, we must combine all of these. To get consistent results at the intersections of L_j's, we must have

$$\phi_j - \phi_k \in 2\pi\hbar\mathbb{Z} \quad \text{on } L_j \cap L_k$$

This condition is equivalent to the so-called **quantization condition**

$$\lambda_L = 2\pi\hbar \times \text{Integral Class}$$

i.e., for all closed loops in L

$$\int \alpha_M \in 2\pi\hbar\mathbb{Z}$$

Example: For the Hamiltonian $H = (p^2 + \omega^2 q^2)/2$, this condition gives

$$E = n\hbar\omega$$

3.3 Handling Non-Projectability

Assume L is non-projectable. The strategy in this case is as follows: Decompose L as $L = \sqcup_j L_j$ where each L_j is projectable, and sum the corresponding contributions:

$$\sum_j (\pi_{L_j}^{-1})^* e^{i\phi_j/\hbar} a$$

Example: In $T^*\mathbb{R} = \{(q,p)\}$ consider the Hamiltonian $H = (p^2 - q)/2$ and $E = 0$. The Lagrangian submanifold is given as

$$L = H^{-1}(0) = \{(x^2, x) \mid x \in \mathbb{R}\} \simeq \mathbb{R}$$

This Lagrangian is clearly non-projectable, but it consists of projectable components

$$L_\pm = \{(x^2, x) \mid x \in \mathbb{R}^\pm\}$$

Let me first determine what $a(x)$ should be. By direct computation, $X_H|_L \sim \partial_x$, so we have $a(x) = A \in \mathbb{R}$. Now, $\alpha_M|_L$ can be evaluated as

$$\alpha_M|_L = 2x^2 dx \Rightarrow \phi = \frac{2x^3}{3}$$

Summing the contributions from $L_\pm$ gives

$$\psi(x) \sim x^{1/4} \cos\left(2x^{3/2}/3\hbar\right)$$

3.4 The Maslov Correction

We start by considering a Lagrangian submanifold $L \subseteq T^*\mathbb{R} \simeq \mathbb{R}^2$. Consider the projection of L to $p-$axis:

$$\pi_p : \mathbb{R}^2 \to p\text{-axis}$$

If the map π_p is a diffeomorphism, then L is said to be p-projectable.

For a p-projectable Lagrangian, we define the function $\tau : L \to \mathbb{R}$ such that its differential yields the symplectic potential in the p-representation:

$$\tau : L \to \mathbb{R} \quad \text{with} \quad d\tau = -q dp|_L$$

Example: Let's consider a simple example where the position is constant. Let

$L = \{(q_0, x) \mid x \in \mathbb{R}\}$

In this case, the position is fixed at $q = q_0$ while momentum $p = x$ varies. For a constant half-density a defined on L, we physically expect the wave function in the position representation to be highly localized around q_0, yielding a Dirac delta function $\delta(q - q_0)$.

Now, by direct computation we have

$$\tau = -q_0 x$$

Pulling this back via the inverse projection, we get

$$(\pi_p^{-1})^* e^{i\tau/\hbar} a = \underbrace{e^{-iq_0 p/\hbar} a|dp|^{1/2}}_{\text{Fourier transform of } \delta(q-q_0)}$$

This precisely matches the expected momentum representation of a delta function localized at q_0.

Now, let's generalize this. For any p-projectable $L \subseteq T^*\mathbb{R}$ with $d\tau = -q dp|_L$, we define

$$B|dp|^{1/2} = (\pi_p^{-1})^* e^{i\tau/\hbar} a$$

The quantum state associated with this Lagrangian submanifold is then given by the inverse semiclassical Fourier transform of B:

$$J_\hbar(L, \iota, \tau, a) := \mathcal{F}_\hbar^{-1}(B)|dq|^{1/2}$$

Let's compare this p-projectable approach with our earlier method that relied on q-projectability.

Example: Consider $L = \{(x, kx) \mid x \in \mathbb{R}\}$. Assume a is a constant half-density on L.
Using the q-representation:

$$pdq \to \frac{kx^2}{2}$$

$$(\pi_L^{-1})^* a = A|dq|^{1/2}$$

This leads to the state:

$$I_\hbar = e^{ikq^2/2\hbar} A|dq|^{1/2}$$

Using the p-representation:

$$-qdp \to -\frac{kx^2}{2}$$

$$(\pi_p^{-1})^* a = |k|^{-1/2} A|dp|^{1/2}$$

We evaluate the generating function in the p-coordinates:

$$\pi_p^* \left(-\frac{kx^2}{2} \right) = -\frac{p^2}{2k}$$

Applying the semiclassical Fourier transform, we find the relation between the two states:

$$J_\hbar = e^{-i\pi \operatorname{sgn}(k)/4} I_\hbar$$

Notice the emergence of the phase factor $e^{-i\pi \operatorname{sgn}(k)/4}$, which is intimately related to the Maslov index.

3.5 Arbitrary Biprojectable Lagrangian Embeddings

Now, consider an arbitrary Lagrangian submanifold $L \subseteq T^*\mathbb{R}$ that is biprojectable, meaning it can be projected onto both the q and p axes locally.

We relate the corresponding functions for the two projections. Let ϕ be the generating function for the q-projection and τ for the p-projection:

$$\phi = \tau + \iota^*(qp)$$

$$S = \phi \circ \pi_L^{-1} \qquad T = \tau \circ \pi_p^{-1}$$

From this Legendre transformation structure, we can express $S(q)$ in terms of $T(p)$:

$$S(q) = -p(q)T'(p(q)) + T(p(q))$$

where the coordinate transformations are governed by:

$$p(q) = S'(q) \longleftrightarrow q = -T'(p)$$

Differentiating the relation $q = -T'(p(q))$ with respect to q, we obtain the relation between the second derivatives:

$$1 = -T''(p(q)) \cdot p'(q) = -T''(p(q)) \cdot S''(q)$$

$$T''(p(q)) = -(S''(q))^{-1}$$

We have two pullbacks for the half-density a:

$$(\pi_L^{-1})^* a = A|dq|^{1/2}$$

$$(\pi_p^{-1})^* a = B|dp|^{1/2}$$

The amplitudes are related by the Jacobian of the transformation between p and q:

$$A(q) = \left|\frac{dp}{dq}\right|^{1/2} B(p(q)) = |S''(q)|^{1/2} B(p(q))$$

The state $J_\hbar$ is given by the integral:

$$J_\hbar = (2\pi\hbar)^{-1/2} \int_{\mathbb{R}} e^{i(pq+T(p))/\hbar} B(p) dp\, |dq|^{1/2}$$

We evaluate this integral using the stationary phase approximation. The phase is $\Phi(p) = pq + T(p)$, and the stationary points occur where $\Phi'(p) = q + T'(p) = 0$, which is consistent with our coordinate relation $q = -T'(p)$. The second derivative of the phase at the stationary point is $T''(p(q))$.

Applying the stationary phase formula yields:

$$J_\hbar = e^{-i\pi \operatorname{sgn}(T''(p(q)))/4} e^{iS(q)/\hbar} |T''(p(q))|^{-1/2} B(p(q)) + O(\hbar)$$

Recognizing the leading term as our previously defined state $I_\hbar$, we establish the connection:

$$J_\hbar = e^{-i\pi \operatorname{sgn}(T''(p(q)))/4} I_\hbar + O(\hbar)$$

Example: Consider $L = \{(x^2, x)\}$ as before.

First, we find

$$d\tau = \iota^*(-qdp) = -(x^2)dx \implies \tau(x) = -\frac{x^3}{3}$$

$$d\phi = \iota^*(pdq) = x(2xdx) = 2x^2dx \implies \phi(x) = \frac{2x^3}{3}$$

The function $T(p)$ is straightforward since $p = x$:

$$T(p) = -\frac{p^3}{3}$$

The relation between q and p is $q = -T'(p) = p^2$. Notice that for $q > 0$, there are two branches of momentum: $p = \pm q^{1/2}$.

The second derivative is:

$$T''(p(q)) = -2p = -2\operatorname{sgn}(p)q^{1/2}$$

Because there are two stationary points (for $p > 0$ and $p < 0$) contributing to a given $q > 0$, the integral expression for $J_\hbar$ becomes a sum of two terms:

$$J_\hbar = \left(\underbrace{e^{-i\pi/4} e^{2iq^{3/2}/3\hbar} B(-q^{1/2})}_{p<0} + \underbrace{e^{i\pi/4} e^{-2iq^{3/2}/3\hbar} B(q^{1/2})}_{p>0} \right) q^{-1/4} |dq|^{1/2}$$

It is crucial to note that while the semiclassical expression diverges at the caustic point $q = 0$ (due to the $q^{-1/4}$ factor), the exact state $J_\hbar$, defined as an integral expression, remains smooth around $q = 0$.

$$J_\hbar \to \text{As an integral expression, smooth around } q = 0$$

With this new method of Maslov, our quantization condition becomes

$$\int_\gamma \alpha_M = 2\pi\hbar(n + \mu_L(\gamma)/4)$$